Math 110 — Week 7

Solutions

HW7 P1.

Idea. $A^2 = -I$ and $(B-I)^2 = 0$ give low-degree polynomials that kill A and B; then rule out degree 1.

(a) $A^2 = -I$, so $m_A \mid x^2+1$. A is not a multiple of I ($Ae_1 = e_2$ is not a multiple of e_1), so $\deg m_A \neq 1$. Hence $m_A = x^2+1$ over $\mathbb{R}$. Over $\mathbb{C}$ the definition (first k with $A^k \in \operatorname{span}(I, \dots, A^{k-1})$) gives the same polynomial: $x^2+1 = (x-i)(x+i)$.

(b) Since

$$(B-I)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = 0$$

and $B \neq I$, $m_B = (x-1)^2$.

(c) A over $\mathbb{R}$: x^2+1 has no real roots, so it is not a product of real linear factors: not diagonalizable. Over $\mathbb{C}$: distinct roots $\pm i$, diagonalizable. B: repeated root, not diagonalizable. This matches Homework 6.

HW7 P2.

Idea. D lowers degree, so a power of it is 0; find the first one.

(a) $D^4 = 0$ on $\mathcal{P}_3(\mathbb{R})$, while $D^3 x^3 = 6 \neq 0$. So $m_D \mid x^4$ and is not x^k for $k \leq 3$: $m_D = x^4$. (b) Repeated root: not diagonalizable. (Also $E(0, D) =$ constants, of dimension $1 < 4$.)

HW7 P3.

(a) **Idea.** If $a_0 = 0$, then x divides m_T; since T is invertible, the factor T can be cancelled, leaving a smaller polynomial that kills T.

If $a_0 = 0$, then $m_T = xq$ with $\deg q = d-1$, and $0 = m_T(T) = Tq(T)$. Multiply by T^{-1}: $q(T) = 0$, contradicting minimality of m_T (if $d = 1$, $q = 1$ and $q(T) = I \neq 0$).

(b) **Idea.** Move the constant term to one side and factor T out of the rest.

From $m_T(T) = 0$: $T\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big) = -a_0 I$. So $T^{-1} = -\frac{1}{a_0}\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big)$.

(c) **Idea.** Apply (b), then factor m_T to read off the eigenvalues.

$a_0 = 4$, so $T^{-1} = -\frac{1}{4}(T^3 + T^2 - T - 5I)$. $m_T = (x-1)^2(x^2+3x+4)$ (check: 1 is a root of m_T and of $m_T' = 4x^3 + 3x^2 - 2x - 5$). The eigenvalues are 1 and $\frac{-3 \pm i\sqrt{7}}{2}$. The root 1 is repeated, so T is not diagonalizable.

HW7 P4.

Idea. Substituting $x \mapsto x + \lambda$ (or $x \mapsto x/c$) turns polynomials that kill T into polynomials that kill $T - \lambda I$ (or cT), and back. So the two minimal polynomials divide each other's transforms and have the same degree.

(a) $p(x) = m_T(x+\lambda)$ is monic of degree $d = \deg m_T$ and $p(T - \lambda I) = m_T(T) = 0$, so $m_{T-\lambda I} \mid p$ and $\deg m_{T-\lambda I} \leq d$. Likewise $r(x) = m_{T-\lambda I}(x-\lambda)$ is monic with $r(T) = m_{T-\lambda I}(T - \lambda I) = 0$, so $d \leq \deg m_{T-\lambda I}$. A monic divisor of p of the same degree is p.

(b) $p(x) = c^d m_T(x/c)$ is monic of degree d (leading coefficient $c^d \cdot c^{-d}$) and $p(cT) = c^d m_T(T) = 0$, so $m_{cT} \mid p$. The same argument with cT and $1/c$ gives $d \leq \deg m_{cT}$. So $m_{cT} = p$.

HW7 P5.

Idea. The roots of m_A are the eigenvalues and $\deg m_A \le 2$, which leaves two candidates.

$\deg m_A \le 2$ (Cayley–Hamilton) and its only root is λ, and over $\mathbb{C}$ it splits, so $m_A = x-\lambda$ or $(x-\lambda)^2$. $m_A = x-\lambda$ exactly when $A - \lambda I = 0$. A is diagonalizable iff m_A has no repeated root iff $m_A = x-\lambda$ iff $A = \lambda I$.

HW7 P6.

Idea. Triangular, so the eigenvalues are on the diagonal; list the candidate minimal polynomials and test them.

Eigenvalues $2, 3$.

$$E(2) = \operatorname{null}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \operatorname{span}(e_1),$$
$$E(3) = \operatorname{null}\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \operatorname{span}(e_3).$$

Candidates (roots exactly $2, 3$, degree ≤ 3): $(x-2)(x-3)$, $(x-2)^2(x-3)$, $(x-2)(x-3)^2$. $(C-2I)(C-3I)$ and $(C-2I)(C-3I)^2$ have $(1,2)$ entry $\mp 1 \neq 0$; $(C-2I)^2 = \operatorname{diag}(0,0,1)$ and $(C-2I)^2(C-3I) = 0$. So $m_C = (x-2)^2(x-3)$: not diagonalizable ($\dim E(2) + \dim E(3) = 2 < 3$).

HW7 P7.

Idea. Lagrange interpolation: a product of factors $(x-\lambda_k)$ vanishes at every λ_k but λ_j; divide by its value at λ_j.

$$p_j(x) = \prod_{k \neq j} \frac{x-\lambda_k}{\lambda_j - \lambda_k}.$$

Since m_T has distinct roots, T is diagonalizable and $V = \bigoplus_k E(\lambda_k, T)$. For $v \in E(\lambda_k, T)$, $p_j(T)v = p_j(\lambda_k)v$, which is v if $k = j$ and 0 otherwise. So for $v = \sum_k v_k$, $p_j(T)v = v_j$: the projection onto $E(\lambda_j, T)$ along $\bigoplus_{k \neq j} E(\lambda_k, T)$.

M1.

Idea. Find a low-degree polynomial relation, then rule out its proper divisors. In (c), B is block diagonal, so m_B is the lcm of the blocks' minimal polynomials (F4).

(a) Every entry of A^2 is 3, so $A^2 = 3A$ and $m_A \mid x(x-3)$. $A \neq 0$ and $A \neq 3I$, so $m_A = x(x-3)$. Distinct roots: diagonalizable.

(b) $Pe_1 = e_2$, $Pe_2 = e_3$, $Pe_3 = e_1$, so $P^3 = I$ and $m_P \mid x^3 - 1$. Since e_1, Pe_1, P^2e_1 is a basis, no nonzero polynomial of degree ≤ 2 kills e_1, so $m_P = x^3 - 1 = (x-1)(x^2+x+1)$. Over $\mathbb{C}$ the roots $1, \omega, \bar{\omega}$ are distinct: diagonalizable. Over $\mathbb{R}$, x^2+x+1 has no real root: not diagonalizable.

(c) B is block diagonal with blocks $B_1 = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, (3), (5). $B_1 \neq 3I$ and $(B_1 - 3I)^2 = 0$, so $m_{B_1} = (x-3)^2$; the other blocks give $x-3$ and $x-5$. So

$$m_B = \operatorname{lcm}\big((x-3)^2, x-3, x-5\big) = (x-3)^2(x-5).$$

Repeated root: not diagonalizable.

M2.

(a) **Idea.** Test the basis vectors: x^k is an eigenvector.

$T(x^k) = kx^k$, so $1, x, \dots, x^n$ is an eigenbasis with distinct eigenvalues $0, 1, \dots, n$. T is diagonalizable and $m_T = x(x-1)\cdots(x-n)$.

(b) **Idea.** $S - I$ is a "difference operator" that lowers degree, like differentiation; so it is nilpotent, and the question is the first power that kills everything.

$(S-I)p = p(x+1) - p(x)$: the leading terms cancel, so deg drops by at least 1 and constants go to 0. Hence $(S-I)^{n+1} = 0$, so $m_S \mid (x-1)^{n+1}$. Also

$$(S-I)x^k = (x+1)^k - x^k = kx^{k-1} + (\text{lower}),$$

so $(S-I)^n x^n = n! \neq 0$. So $m_S = (x-1)^{n+1}$, a repeated root: not diagonalizable.

M3.

Idea. $T^3 = T$ says $x^3 - x$ kills T; factor it.

$m_T \mid x^3 - x = x(x-1)(x+1)$, so m_T is a product of distinct real linear factors and T is diagonalizable. Its eigenvalues are roots of m_T, hence in $\{-1, 0, 1\}$.

M4.

Idea. A monic polynomial of degree 1 is $x - c$, and $m_T = x - c$ says $T - cI = 0$.

If $T = cI$, then $x - c$ kills T and $\deg m_T \geq 1$ (as $V \neq 0$), so $m_T = x - c$. Conversely, if $m_T = x - c$, then $T - cI = m_T(T) = 0$.

M5.

Idea. Substitute $x \mapsto 1/x$ in $m_T(T) = 0$ and clear denominators by multiplying by T^{-d}.

$$q(x) = \frac{1}{a_0} x^d m_T(1/x) = \frac{1}{a_0}\left(1 + a_{d-1}x + \cdots + a_0 x^d\right)$$

is monic of degree d ($a_0 \neq 0$ by HW7 P3). $q(T^{-1}) = \frac{1}{a_0} T^{-d} m_T(T) = 0$, so $m_{T^{-1}} \mid q$ and $\deg m_{T^{-1}} \leq d$. Applying this to T^{-1} in place of T gives $d \leq \deg m_{T^{-1}}$. So $m_{T^{-1}} = q$.

M9.

Idea. $m_A(A) = 0$ is a rule for rewriting A^2 in terms of A and I; use it to reduce any power, and solve it for A^{-1}.

A is not scalar, so $m_A = \chi_A = x^2 - (\operatorname{tr} A)x + \det A = x^2 - 5x - 2$. From $A^2 - 5A = 2I$: $A(A - 5I) = 2I$, so

$$A^{-1} = \frac{1}{2}(A - 5I) = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}.$$

Multiply $A^2 = 5A + 2I$ by A and replace A^2 again:

$$A^3 = A(5A + 2I) = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I.$$